# 1 Multiplicative Function

**Definition 1.1** (Multiplicative Function). *A number-theoretic function $f$ is* ***multiplicative*** *if $f(1) = 1$ and $f(mn) = f(m)f(n)$, $\forall m, n \in \mathbb{N}$ such that $gcd(m, n) = 1$. Additionally, $f$ is called* ***completely multiplicative*** *if $f(mn) = f(m)f(n)$, $\forall m, n \in \mathbb{N}$.*

## Examples.

- $1(n) = 1$: The constant function.
- $Id(n) = n$: The identity function.
- $\varepsilon(n) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } n > 1 \end{cases}$ : The unit function
- $\tau(n) = \sum_{d|n} 1$: The number of divisors function.
- $\sigma(n) = \sum_{d|n} d$: The sum of divisors function.
- $\phi(n) = \sum_{i=1}^{n} [gcd(i, n) = 1]$: Euler's Totient Function (here the third brackets serve as a boolean function, which returns 1 if the condition is true, 0 otherwise.)

Note that $1, Id, \varepsilon$ are completely multiplicative as well, while $\phi, \tau, \sigma$ aren't.